

Essential Summary

Quantitative Analysis of Dynamical Bifurcations in a Coupled Smooth and Discontinuous Oscillator with High-order Nonlinear Damping

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1 Background and Motivation

The coupled smooth and discontinuous (SD) oscillator, introduced by Han et al. (2012), consists of two SD oscillators coupled rigidly. Its restoring force contains **two irrational nonlinear terms**, which arise from the geometric configuration of oblique springs. These irrational terms make the system significantly harder to analyze than classical nonlinear oscillators.

When perturbed by high-order nonlinear damping of the form $(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x}$, the oscillator exhibits rich dynamical behavior: limit cycles of varying stability, homoclinic and heteroclinic bifurcations, and multi-stable configurations (single-well, double-well, and triple-well potentials).

Classical quantitative methods—the elliptic function L–P method, the elliptic perturbation method, and the standard GHFP method—**cannot handle** this system because the two irrational terms render the key integrals analytically intractable. This work overcomes that barrier by extending the **modified generalized harmonic function perturbation (MGHFP)** method, originally developed by the authors for simpler oscillator classes, to this most challenging case.

2 System Model

The investigated oscillator is governed by:

$$\ddot{x} + (x + \beta) \frac{1}{\sqrt{(x + \beta)^2 + \alpha^2}} + (x - \beta) \frac{1}{\sqrt{(x - \beta)^2 + \alpha^2}} = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x} \quad (1)$$

where α is the smoothing parameter ($\alpha \rightarrow 0$ gives the non-smooth/discontinuous limit), β is the coupling parameter controlling the potential well structure, ε is a small positive perturbation parameter, μ_c is the control parameter, and μ_1 – μ_4 are nonlinear damping coefficients.

The potential energy landscape undergoes qualitative changes as α and β vary: monostable (single-well) for large α/β ratios, bistable (double-well) at intermediate values, and tristable (triple-well) for small α with appropriate β .

3 The MGHFP Method — Key Steps

The method proceeds in three stages:

3.1 Stage 1: Nonlinear Time Transformation

Introduce a nonlinear time transformation $\frac{d\varphi}{dt} = \Phi(\varphi)$ with $\varphi(0) = 0$, $\varphi(t + \tau) = \varphi(t) + \pi$. The undamped system is transformed into:

$$\frac{d}{d\varphi}(\Phi x') + g(x) = 0, \quad x(\varphi) = a \cos^2 \Phi(\varphi) + b \quad (2)$$

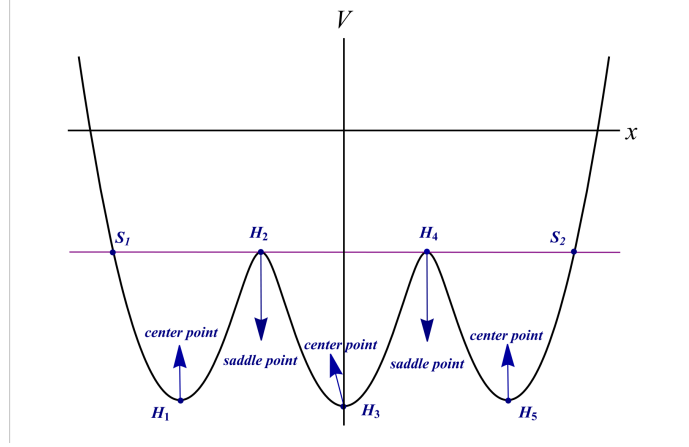


Figure 1: **Triple-well potential** of the coupled SD oscillator. The system has five equilibrium points: two centers, two saddles, and an additional center, creating a complex landscape where both homoclinic and heteroclinic orbits can coexist.

where $g(x) = (x + \beta) \left[1 - \frac{\alpha^2}{(x+\beta)^2 + \alpha^2} \right] + (x - \beta) \left[1 - \frac{\alpha^2}{(x-\beta)^2 + \alpha^2} \right]$.

The function $\Phi(\varphi)$ is expanded as a Fourier series:

$$\Phi_0(\varphi) = \sum_{i=0}^m (p_{2i} \cos 2i\varphi + q_{2i} \sin 2i\varphi) \quad (3)$$

3.2 Stage 2: Perturbation Expansion

The amplitude and frequency are expanded in powers of ε :

$$a = a_0 + \varepsilon a_1 + \dots, \quad \Phi = \Phi_0 + \varepsilon \Phi_1 + \dots \quad (4)$$

Substituting into the damped system and collecting terms at each order of ε yields a hierarchy of equations. The zeroth-order gives the unperturbed solution; the first-order determines a_1 and Φ_1 .

3.3 Stage 3: Composite Simpson Integration — The Key Innovation

The Fourier coefficients p_{2i} involve integrals of the irrational nonlinear terms that **cannot be evaluated analytically**. The MGHFP method's key innovation is the introduction of the **composite Simpson quadrature formula** to discretize these integrals into sums over equally-spaced nodes, making the coefficients computable to high precision.

Specifically, the integration interval $[0, \pi]$ is divided into eight equal parts, and the Simpson formula yields explicit expressions for each p_{2i} in terms of a_0 , b , α , and β (listed in Appendix A of the original paper).

4 Key Equations

4.1 Amplitude–Parameter Relation (Central Result)

Setting the Fourier truncation order $m = 4$, the analytical relationship between the limit cycle amplitude A and the control parameter μ_c is:

$$\mu_c = -\frac{4}{a_0^2(2p_0 - p_4)} \left(E_1\mu_1 + E_2\mu_2 + E_3\mu_3 + E_4\mu_4 \right) \quad (5)$$

where p_{2i} are the Fourier coefficients computed via composite Simpson integration, and E_1-E_4 are explicit functions of a_0 , b , and p_{2i} . **This is the central result:** given system parameters, the limit cycle amplitude is obtained directly—no numerical integration needed.

For symmetric limit cycles: $a_0 = 2A$, $b = -A$. For asymmetric limit cycles (triple-well case): $a_0 = A - C$, $b = C$, where C depends on α and β .

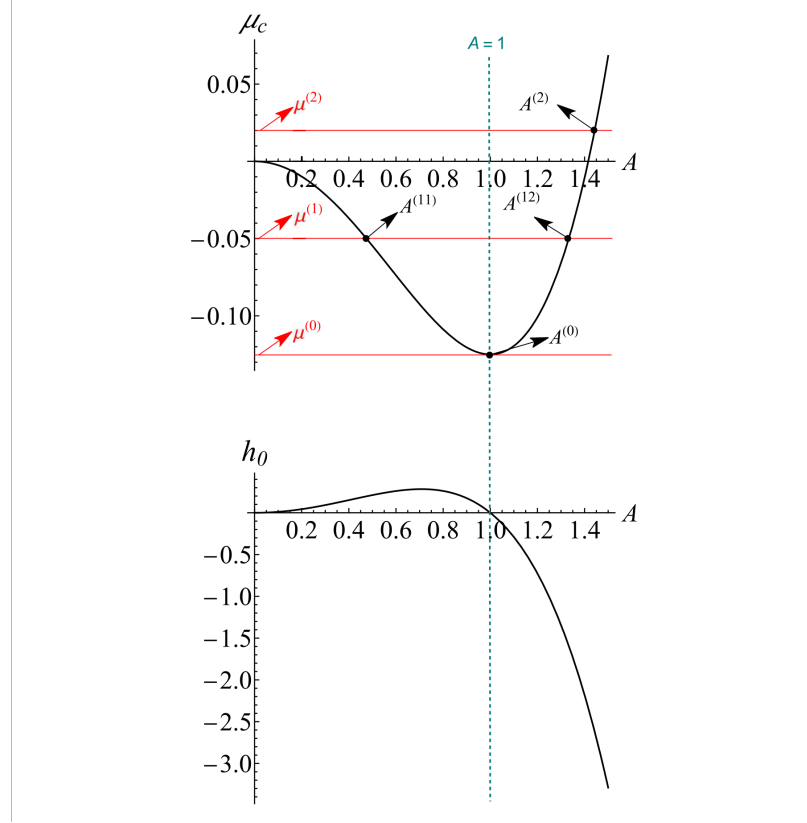


Figure 2: **Figure 1.** (Upper) Relationship between limit cycle amplitude A and control parameter μ_c for the single-well oscillator ($\alpha = 2$, $\beta = 0.5$). (Lower) Characteristic quantity h_0 determining stability. Key bifurcation points: $\mu^{(0)} = -0.1249$ (semi-stable cycle emerges), $\mu^{(1)} = -0.05$ (split into stable + unstable), $\mu^{(2)} = 0.02$ (unstable cycle collapses). The complete lifecycle of limit cycles is quantitatively captured.

4.2 Stability Criterion

The stability of each limit cycle is determined by the characteristic quantity:

$$h_0 = \frac{1}{\pi} \int_0^\pi \left[\mu_c + \sum_{i=1}^4 \mu_i (a_0 \cos^2 \varphi + b)^i \right] d\varphi \quad (6)$$

Based on the qualitative theory of ODEs (Dumortier, Llibre & Artés, 2006):

- $h_0 > 0$: the limit cycle is **unstable**
- $h_0 = 0$: the limit cycle is **semi-stable** (a bifurcation point)
- $h_0 < 0$: the limit cycle is **stable**

4.3 Homo-heteroclinic Bifurcation Thresholds

Setting $b = h$ (saddle point ordinate) in Eq. (5) yields the critical parameter μ_c^{hom} for **homoclinic** bifurcation. Setting $b = h_2$ and $a_0 + b = h_1$ (connecting two saddle points) yields μ_c^{het} for **heteroclinic** bifurcation. For the triple-well system, both can coexist at the same saddle—a scenario uniquely handled by this method.

4.4 First-Order Approximate Solution

The limit cycle and its derivative are approximated as:

$$x = 2A \cos^2 \Phi - A + \mathcal{O}(\varepsilon^2) \quad (7)$$

$$\dot{x} = -2a_0[\Phi_0 + \varepsilon\Phi_1] \cos \varphi \sin \varphi + \mathcal{O}(\varepsilon^2) \quad (8)$$

where $\Phi_0 = p_0 + p_2 \cos 2\varphi + p_4 \cos 4\varphi + p_6 \cos 6\varphi + p_8 \cos 8\varphi + \dots$ and $\Phi_1 = l_2 \sin 2\varphi + l_4 \sin 4\varphi + l_6 \sin 6\varphi + l_8 \sin 8\varphi + \dots$

5 Validated Examples

5.1 Example 1: Single-Well Potential ($\alpha = 2, \beta = 0.5$)

With $\mu_1 = 1, \mu_2 = 1, \mu_3 = 1, \mu_4 = -1$, the amplitude–parameter curve (Figure 1) reveals:

- $\mu_c < \mu^{(0)} = -0.1249$: **no limit cycle exists**.
- At $\mu_c = \mu^{(0)}$: a **semi-stable** limit cycle emerges (outer-stable, inner-unstable).
- $\mu^{(0)} < \mu_c < 0$: the semi-stable cycle **splits into two**—a smaller unstable cycle ($h_0 > 0$) and a larger stable cycle ($h_0 < 0$).
- At $\mu_c = 0$: the unstable cycle **collapses** to the singular point.
- $\mu_c > 0$: only a single **stable** limit cycle persists, growing with μ_c .

Accuracy metrics (Maximum Absolute Error in Amplitude, RMS Phase Error, and L_2 -norm difference) confirm reliability: when $|\varepsilon\mu_i| \leq 0.5$, all errors remain below 5%.

5.2 Example 2: Triple-Well Potential

The triple-well case is considerably more complex: the system has **five equilibrium points**, and both homoclinic and heteroclinic orbits can coexist. The MGHFP method successfully:

- Predicts the critical values of μ_c for both homoclinic and heteroclinic bifurcations.
- Obtains approximate analytical solutions for the homo-heteroclinic orbits themselves.
- Tracks the evolution of both small limit cycles (enclosed by homoclinic orbits) and large limit cycles (enclosing all equilibrium points).

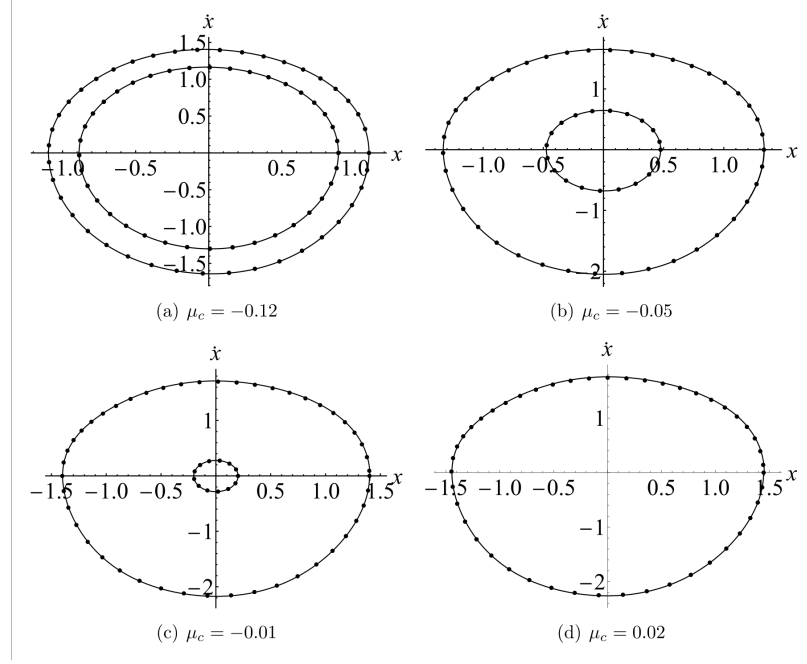


Figure 3: **Figure 2.** Phase portraits of limit cycles for the single-well oscillator at different μ_c values ($\varepsilon = 0.2$). Solid lines: Runge–Kutta numerical integration. Dotted lines: present MGHFP analytical method. The excellent agreement confirms accuracy even at moderately large ε .

6 Key Findings

1. **Full quantitative prediction:** The MGHFP method achieves, for the first time, analytical quantitative analysis of limit cycles in a coupled SD oscillator with two irrational nonlinearities. Given system parameters, one can directly compute the number, amplitude, and existence interval of every limit cycle—no repetitive numerical search needed.
2. **Analytical stability criterion:** The characteristic quantity h_0 distinguishes stable, unstable, and semi-stable limit cycles analytically. Unstable and semi-stable limit cycles, which are exceedingly difficult to capture by purely numerical means, are reliably identified by this method.
3. **Simultaneous homo-heteroclinic bifurcation prediction:** For the triple-well system, homoclinic and heteroclinic orbits coexist at the same saddle. The MGHFP method predicts both bifurcation parameters simultaneously—a capability not available from prior analytical approaches.
4. **Complete limit cycle lifecycle:** The entire lifecycle is quantitatively tracked—from birth (semi-stable bifurcation), through evolution (splitting into stable + unstable pair), to annihilation (shrinking to a singular point).
5. **High-order damping applicability:** With quartic nonlinear damping, the method maintains reliable accuracy when $|\varepsilon\mu_i| \leq 0.5$. Beyond this range, accuracy is recovered by increasing the Fourier truncation order.

7 Method Limitations

The authors identify three conditions where accuracy degrades:

- *Large perturbation parameter* ($\varepsilon > 0.5$): Relative error exceeds 5%. Extending to higher-order perturbation expansions can mitigate this.

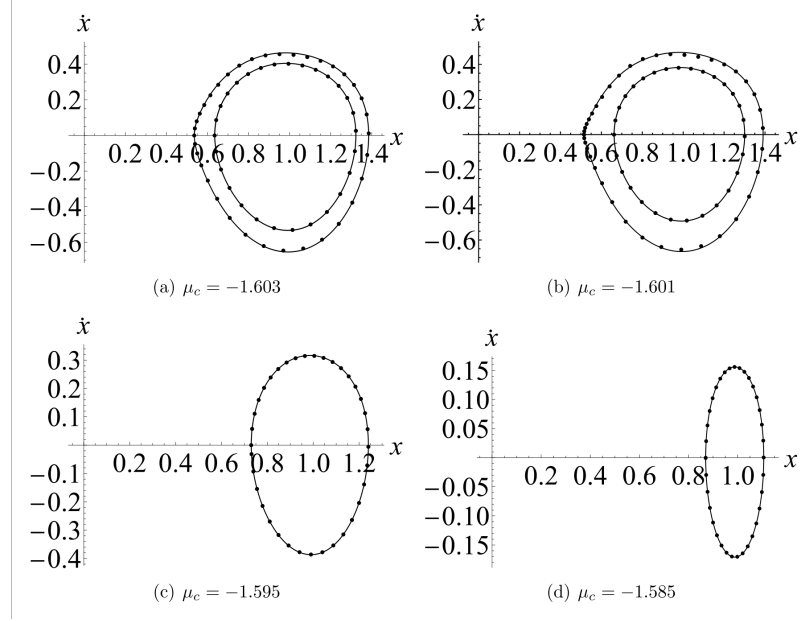


Figure 4: **Figure 7.** Limit cycles for the triple-well oscillator at different μ_c values ($\varepsilon = 1$). Despite the large perturbation parameter, the MGHFP results (dotted) closely match the Runge–Kutta reference (solid), demonstrating the method’s robustness in complex multi-stable systems.

- *Limit cycles crossing saddle points:* The velocity drops sharply near the saddle, causing abrupt displacement waveform features. The Fourier series truncated at $m = 4$ may be insufficient; higher truncation orders are needed.
- *Non-smooth potential wells ($\alpha = 0$):* The non-smoothness affects accuracy. A smaller ε or higher truncation order is recommended.

8 Applications and Significance

The mathematical model investigated—combining irrational nonlinear stiffness, high-order nonlinear damping, and multi-stable dynamics—has direct relevance to two engineering domains:

(1) Quasi-zero stiffness (QZS) vibration isolators: Recent research has extensively explored QZS isolators incorporating nonlinear electromagnetic damping for low-frequency vibration attenuation. The coupled SD oscillator’s irrational stiffness and high-order damping naturally model such systems, and the MGHFP method provides a quantitative tool for predicting their limit cycle behavior and bifurcation thresholds.

(2) MEMS resonators: In micro-electro-mechanical systems, nonlinear damping and multi-stable dynamics are increasingly exploited for performance enhancement. The bifurcation analysis framework developed here directly supports the design and optimization of such devices, where saddle-node bifurcations and multi-stability play critical roles.

Beyond these specific applications, the MGHFP method itself constitutes a versatile analytical tool: it can be transferred to other systems with irrational or non-smooth nonlinearities, wherever classical perturbation methods fail.

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